Adrenal Disorders I: Adrenal Physiology, Conn's Syndrome & Cushing's Syndrome

Internal Medicine Department | Source: `12-adrenal gland physiology cushing syndrome.pdf` (44 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **Pituitary I–II:** ACTH from basophils, CRH control, and trans-sphenoidal surgery. **Cushing disease = an ACTH-secreting pituitary microadenoma.**
> - **Hypertension:** Conn's and Cushing's were on the **endocrine** list of secondary HTN, and **spironolactone/eplerenone** were the "aldosterone antagonist" drugs.
> - **Lung cancer: SCLC/carcinoid → ectopic ACTH**.
> - **Physiology (hormones):** steroids are lipid-soluble and act on **intracellular receptors**.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and aetiology of adrenal disorders	Zone → hormone; RAAS
Knowledge	⭐ Epidemiology, manifestations, complications, management of **hyper**adrenalism	Conn's (HTN + ↓K), Cushing's (features, tests)
Knowledge	Causes and manifestations of hirsutism	(Lecture 13)
Skill	⭐ Interpret investigations → diagnosis	Aldosterone/renin ratio; dexamethasone tests; ACTH
Skill	Evidence-based management	Adrenalectomy vs MRA; trans-sphenoidal surgery
Skill	Emergencies	Hypokalaemic paralysis; post-op adrenal insufficiency

Main subtopics: anatomy and physiology (histology, steroidogenesis, RAAS) → **primary hyperaldosteronism / Conn's** (causes, picture, investigations, treatment, DDx) → **Cushing's** (causes: pituitary, adrenal, ectopic, exogenous; picture) → **investigations** (screening, confirmation, differentiation, localisation, DDx) → **treatment and Nelson's syndrome**.

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2. Main Content (Lecture Order)

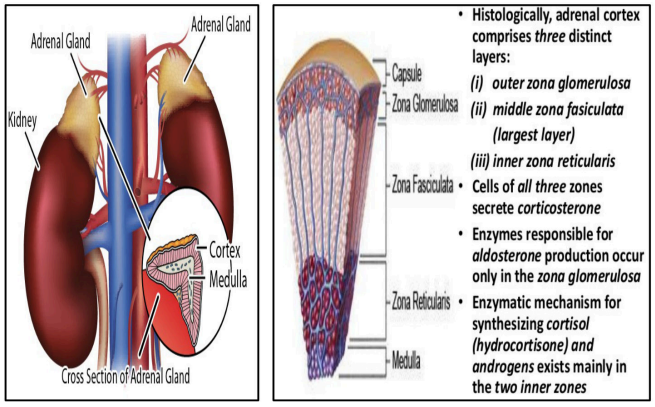
2.1 Anatomy & Histology (Slides 4–6) ⭐

Layer	Hormone	Control
Capsule	—	—
⭐ **Zona Glomerulosa** (outer)	**Mineralocorticoids** (aldosterone)	**RAAS** (angiotensin II), **K^+**

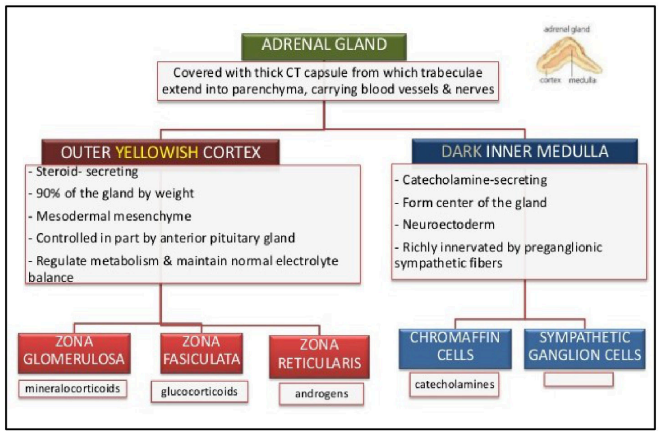
Layer	Hormone	Control
⭐ **Zona Fasciculata** (middle, largest)	**Glucocorticoids** (cortisol)	**ACTH**
⭐ **Zona Reticularis** (inner)	**Androgens** (DHEA)	**ACTH**
Medulla (chromaffin cells, neuroectoderm)	**Catecholamines**	**Preganglionic sympathetic** fibres

> **Mnemonic: "GFR – Salt, Sugar, Sex"**: **G**lomerulosa = salt, **F**asciculata = sugar, **R**eticularis = sex. "The deeper you go, the sweeter it gets."

- **Cortex: mesodermal**, **90% of the gland**, outer yellow layer, steroid-secreting. Partly controlled by the anterior pituitary; regulates metabolism and maintains normal electrolyte balance.
- **Medulla: neuroectodermal**, dark inner layer, richly innervated by **preganglionic sympathetic fibres**.



Slide 5: adrenal anatomy and zones



Slide 6: cortex vs medulla

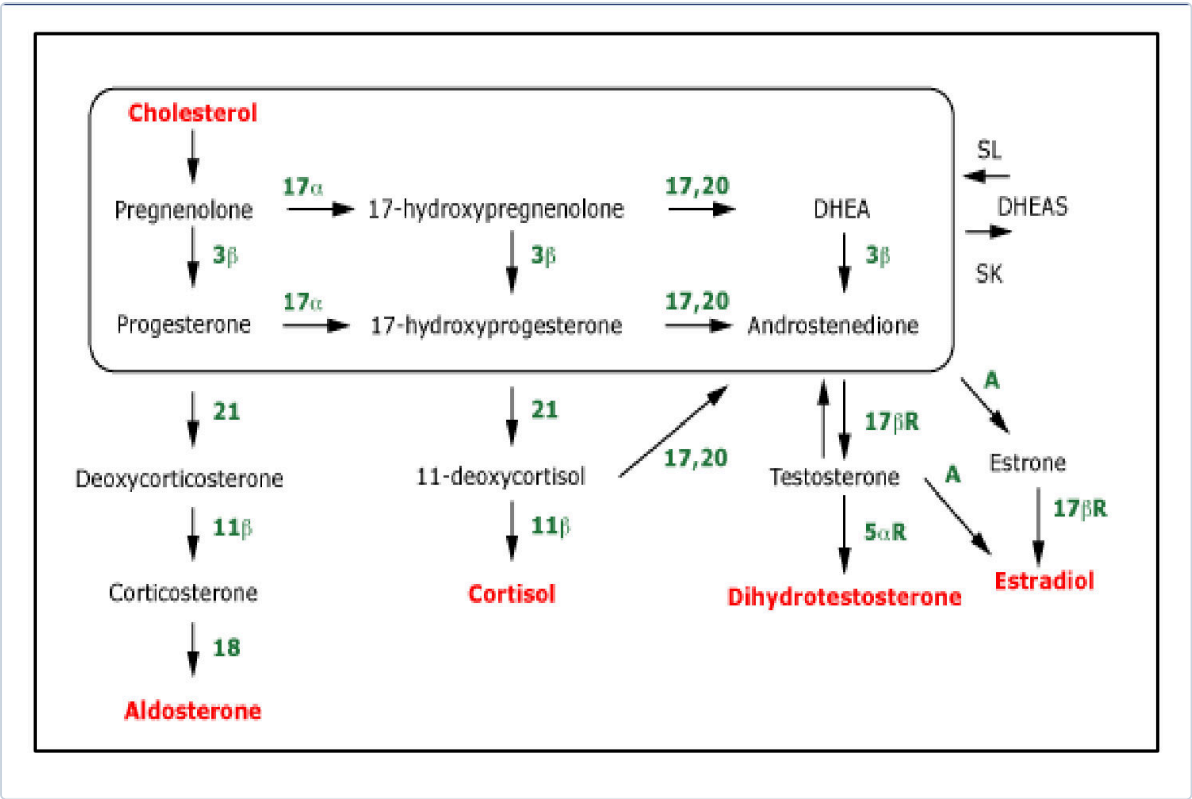
2.2 Steroid Synthesis (Slide 7) ⭐

Cholesterol → pregnenolone → three pathways:

- **Mineralocorticoid:** progesterone →(**21**)→ deoxycorticosterone →(**11β**)→ corticosterone →(**18**)→ **aldosterone**
- **Glucocorticoid:** 17-OH-pregnenolone → 17-OH-progesterone →(**21**)→ 11-deoxycortisol →(**11β**)→ **cortisol**
- **Androgen:** (17,20-lyase) → **DHEA** → androstenedione → **testosterone** →(5α-reductase)→ DHT; aromatase → oestrogens

Key enzymes:

- **17α-hydroxylase** sends steroids to the cortisol and sex-steroid pathways.
- **21-hydroxylase** and **11β-hydroxylase** are needed for **both** aldosterone and cortisol.
- This is the key to the CAH types in lecture 13.

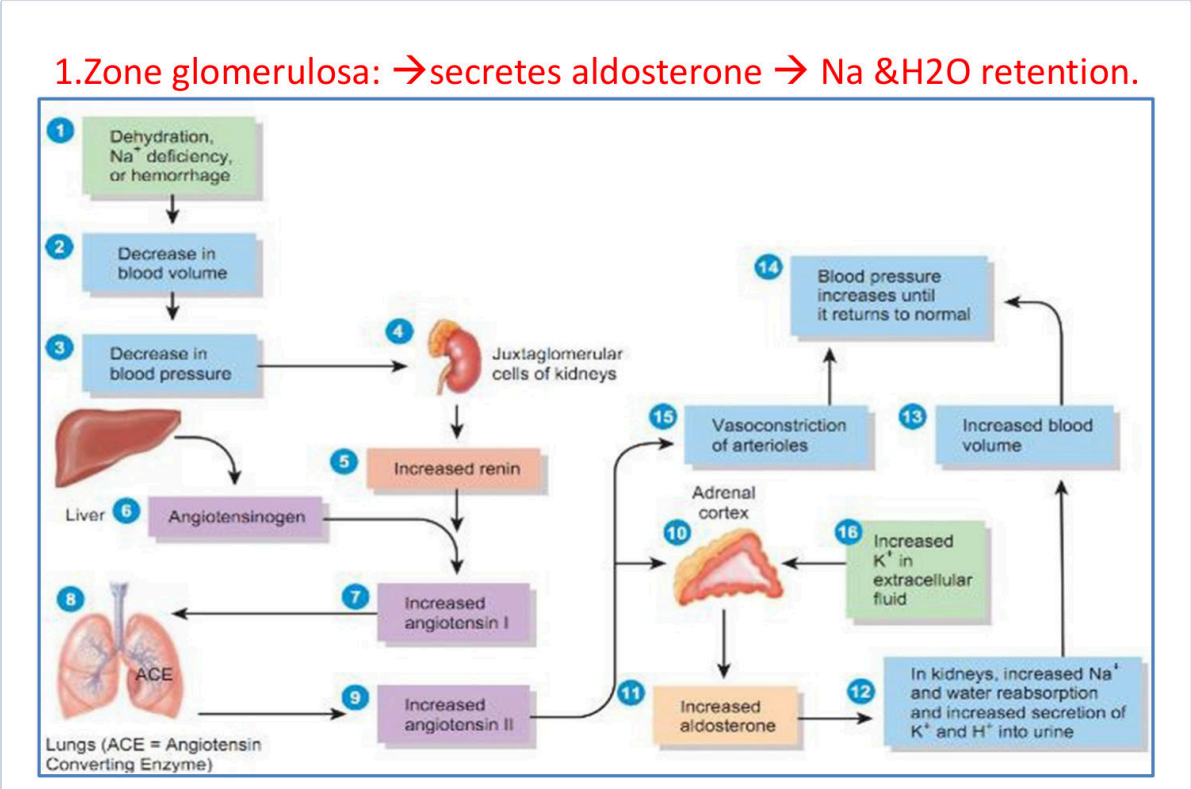


Slide 7: synthesis of adrenal hormones

2.3 Physiology by Zone (Slides 8–10)

1. Zona glomerulosa → aldosterone → Na^+ and water retention (and K^+/H^+ excretion *[added]*).

- **RAAS (slide 8):** ↓ BP or volume → juxtaglomerular renin → angiotensinogen → angiotensin I →(ACE, lungs)→ angiotensin II → vasoconstriction + **aldosterone** → Na/water retention → BP ↑ (negative feedback).



Slide 8: renin–angiotensin–aldosterone system

2. Zona fasciculata → glucocorticoids (cortisol). Metabolic effects (and effects in excess):

Target	Effect
Fat	Mobilisation; **in excess → deposition in abnormal sites**
Protein	**Catabolic**; in excess → **muscle wasting and osteoporosis**
Carbohydrate	↑ **gluconeogenesis**, ↓ muscle glucose uptake; in excess → **hyperglycaemia**
Water and electrolytes	**Na^+ and water retention, K^+ excretion**
Blood (marrow)	↑ **RBCs**; ↓ **eosinophils and lymphocytes**
Other	**Androgenic**, **anti-allergic**, **anti-inflammatory** (in excess ↓ fibrous tissue, destroys elastic fibres)

3. Zona reticularis → sex hormones, mainly androgens (+ a little oestrogen and progesterone).

- Main androgen: **dehydroepiandrosterone (DHEA)**: protein anabolism, male secondary sexual characteristics.
- End products: **17-ketosteroids** (urine).
- Under **ACTH** control.
- **Androgen sources:** ♂ adrenal + testes; ♀ adrenal (*[added]* + ovary).

> 🔁 **Recap: Physiology.** Cortex (mesoderm, 90%): glomerulosa = aldosterone (RAAS, K^+), fasciculata = cortisol (ACTH), reticularis = androgens (ACTH). Medulla (neuroectoderm) = catecholamines. Cortisol: gluconeogenesis, protein catabolism, fat redistribution, Na retention, ↑RBC, ↓eosinophils and lymphocytes, anti-inflammatory.
>
> 🔗 **Link:** Too much of the outermost hormone (aldosterone) gives the commonest curable cause of secondary hypertension.
>
> ▶️ **Up next: Conn's syndrome.**

2.4 Conn's Syndrome (Primary Hyperaldosteronism): Causes (Slides 11–12) ⭐

1. ⭐ **Aldosterone-producing adrenal adenoma: 60%** (Conn's adenoma)
2. **Bilateral zona glomerulosa hyperplasia: 40%**
3. **Aldosterone-producing carcinoma** (rare)

[added] More recent data show bilateral hyperplasia now outnumbers adenoma in screening series. The slide's 60/40 is the classic textbook split.

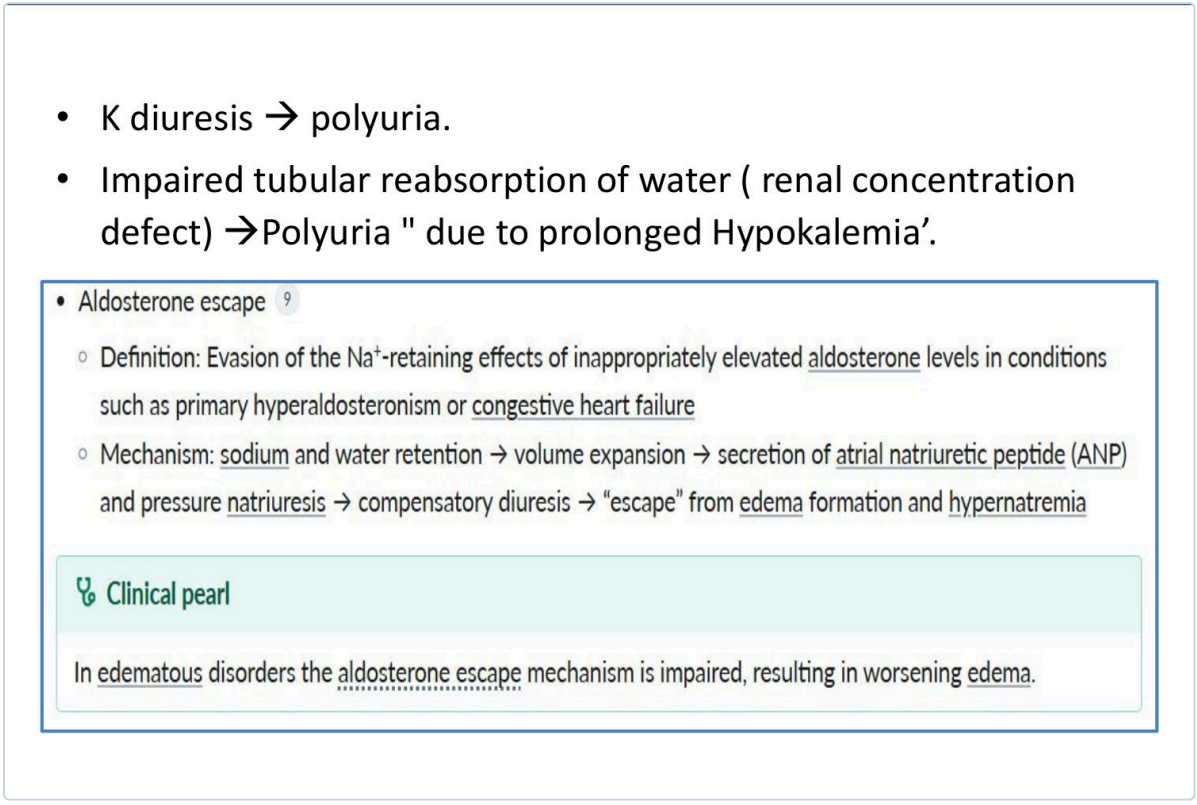
2.5 Clinical Picture (Slides 13–14) ⭐

#	Feature	Detail
1	⭐ **Hypertension**	May be severe (Na retention)
2	⭐ **Hypokalaemia**	**Apathy, paraesthesia**; **arrhythmias** (extrasystoles); **intestinal atony → constipation → paralytic ileus**; **muscle weakness → episodic paralysis**; **impaired glucose tolerance** (K^+ needed for insulin release and action)
3	**Metabolic alkalosis**	Heavy urinary H^+ loss → **tetany**
4	⭐ **NO oedema**	Because of **aldosterone escape**

#	Feature	Detail
5	**Polyuria**	K^+ diuresis + **hypokalaemic nephropathy** (renal concentrating defect)

Aldosterone escape (slide 14):

- **Definition:** escape from the Na-retaining effect of excess aldosterone in conditions such as primary hyperaldosteronism or congestive heart failure.
- **Mechanism:** Na and water retention → ↑ volume → **ANP** secretion and pressure natriuresis → compensatory diuresis → **"escape" from oedema and hypernatraemia**.
- **Pearl:** in **oedematous** disorders (HF, cirrhosis, nephrotic) the escape mechanism is **impaired**, so oedema worsens.



Slide 14: aldosterone escape (polyuria, no oedema)

2.6 Investigations (Slides 15–17, 21) ⭐

Group	Finding
Biochemistry	**Hypokalaemia** (N 3.5–5) · **hypernatraemia** (N 140; *[added]* usually high-normal) · **alkalosis** (↑ HCO_3^-; N 22–30)
Urine	**Polyuria** · **↑ K^+, ↓ Na^+** · **↑ aldosterone excretion** (N 12–50 µg/24 h)
⭐ **Plasma renin**	**↓ in 1ry** (stays low on Na restriction). **↑ in 2ry hyperaldosteronism**. ⭐ **Stop antihypertensives before measuring**
Plasma aldosterone	**High** (N 3–10 ng/dL)
Imaging	**US, adrenal CT**

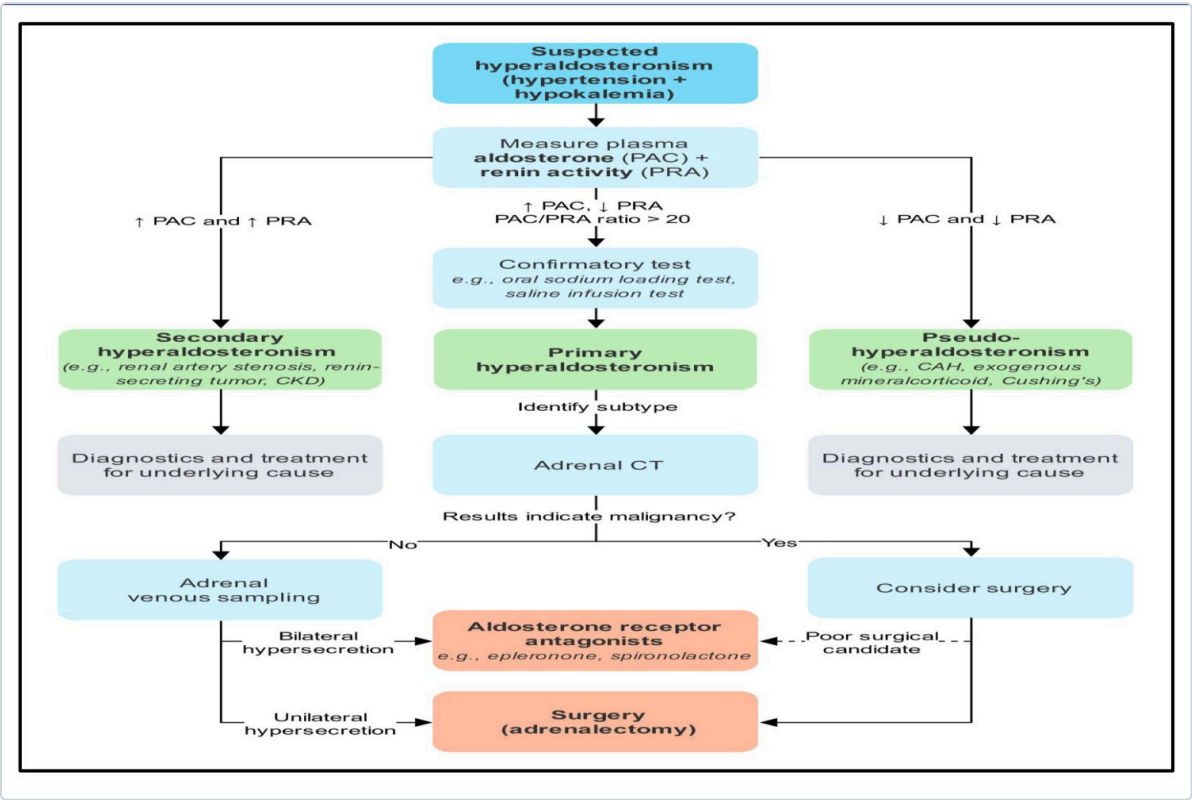
Confirmatory testing (slide 17) shows that aldosterone production is **non-suppressible** (not regulated by RAAS):

- ⭐ **Oral sodium loading:** high Na intake for 3 days, 24-h urine aldosterone on day 3. **> 12 µg/day** = primary hyperaldosteronism highly likely.

- ⭐ **Saline infusion:** baseline renin and aldosterone, NS over 4 h, repeat. **Aldosterone > 10 ng/dL** = very probable.
- **Others: fludrocortisone suppression**, **captopril suppression**.
- **Interpretation:** aldosterone **suppresses** → primary unlikely. **No suppression** → **primary confirmed**.

Diagnostic algorithm (slide 21) ⭐

1. Suspect (HTN + hypokalaemia) → **plasma aldosterone (PAC) + renin activity (PRA)**: - **↑ PAC, ↑ PRA** → **secondary** (renal artery stenosis, renin-secreting tumour, CKD) → treat the cause - **↑ PAC, ↓ PRA, PAC/PRA > 20** → **confirmatory test** → **primary** - **↓ PAC, ↓ PRA** → **pseudo-hyperaldosteronism** (CAH, exogenous mineralocorticoid, Cushing's) → treat the cause
2. Primary → **adrenal CT**: - **malignancy suspected** → surgery - **not malignant** → ⭐ **adrenal venous sampling**: - **unilateral** → **adrenalectomy** - **bilateral** → **aldosterone antagonists** (eplerenone, spironolactone)



Slide 21: hyperaldosteronism diagnostic algorithm

2.7 Treatment (Slide 18)

1. ⭐ **Surgical removal of the adenoma** (*[added]* laparoscopic adrenalectomy), **after correcting the electrolytes**.
2. **Medical** for **bilateral adrenal hyperplasia**: - **Amiloride (40 mg/day)** is preferred to **spironolactone**, because spironolactone causes **gynaecomastia and impotence** with chronic use. - *[added]* **Eplerenone** (selective MRA) avoids the anti-androgen effects.

2.8 Differential Diagnosis (Slides 19–20)

A. Hypertension + hypokalaemia (pseudo-hyperaldosteronism and mimics):

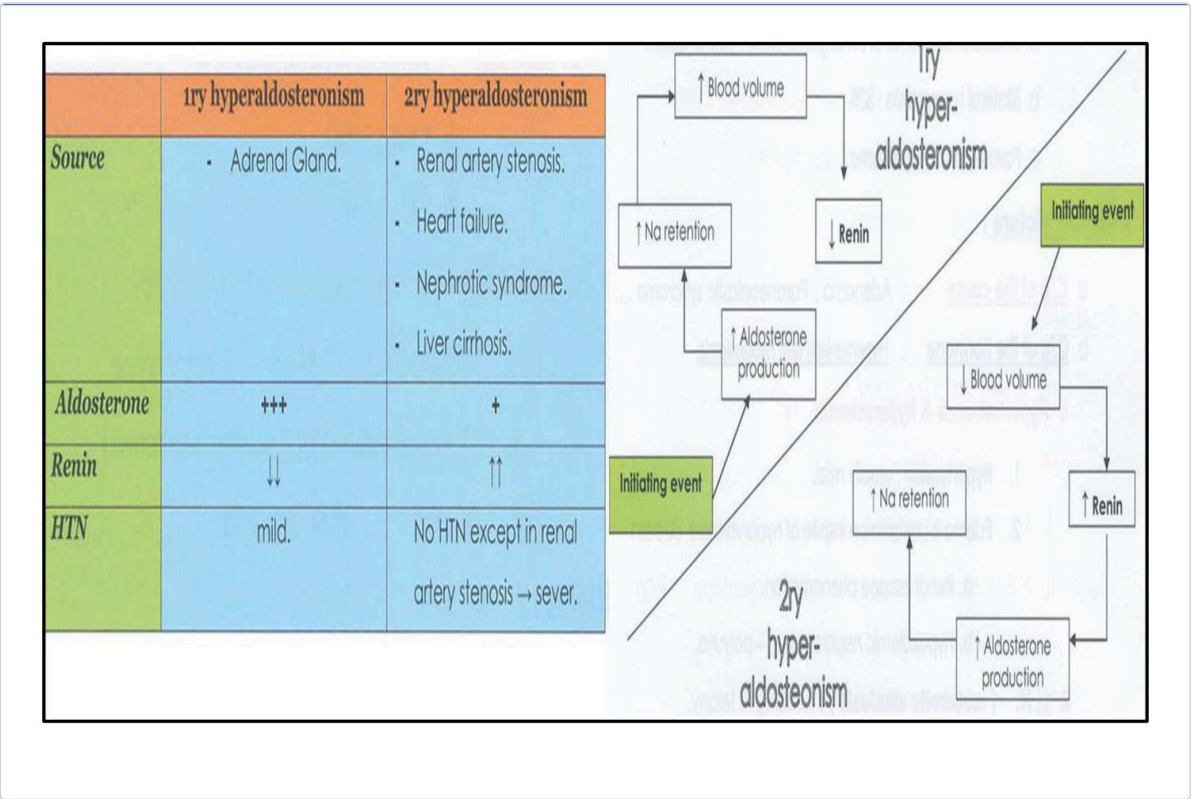
- **Malignant or renal HTN** with **secondary hyperaldosteronism**
- **Diuretic-induced hypokalaemia** in a hypertensive patient
- **Cushing's syndrome**, **CAH** (11β and 17α), rarely **juxtaglomerular hyperplasia**

- **Exogenous mineralocorticoids** (*[added]* liquorice)
- **K-losing nephropathies with HTN**: **Bartter's** (*[added]* actually normotensive), **Liddle's**

B. Other causes of hypokalaemia and **familial periodic paralysis**.

1ry vs 2ry hyperaldosteronism (slide 20):

	1ry	2ry
Source	**Adrenal gland**	**Renal artery stenosis, heart failure, nephrotic syndrome, liver cirrhosis**
Aldosterone	**+++**	+
Renin	⭐ ↓↓	⭐ ↑↑
HTN	Mild (slide)	**No HTN, except renal artery stenosis → severe**
Initiating event	Aldosterone ↑ → Na retention → ↑ volume → ↓ renin	↓ blood volume → ↑ renin → ↑ aldosterone → Na retention



Slide 20: primary vs secondary hyperaldosteronism

🔁 **Recap: Conn's.** Adenoma 60%, bilateral hyperplasia 40%, rarely carcinoma. HTN + hypokalaemia (weakness, paralysis, ileus, arrhythmia, IGT, polyuria) + metabolic alkalosis (tetany), with **no oedema** (aldosterone escape). Labs: ↑ aldosterone, **↓ renin** (stop antihypertensives first), ARR > 20, then confirm non-suppression (oral salt load > 12 µg/day, saline infusion > 10 ng/dL). CT → adrenal venous sampling → unilateral: adrenalectomy; bilateral: MRA/amiloride.

🔗 **Link:** The middle zone's hormone in excess gives a far more visible syndrome.

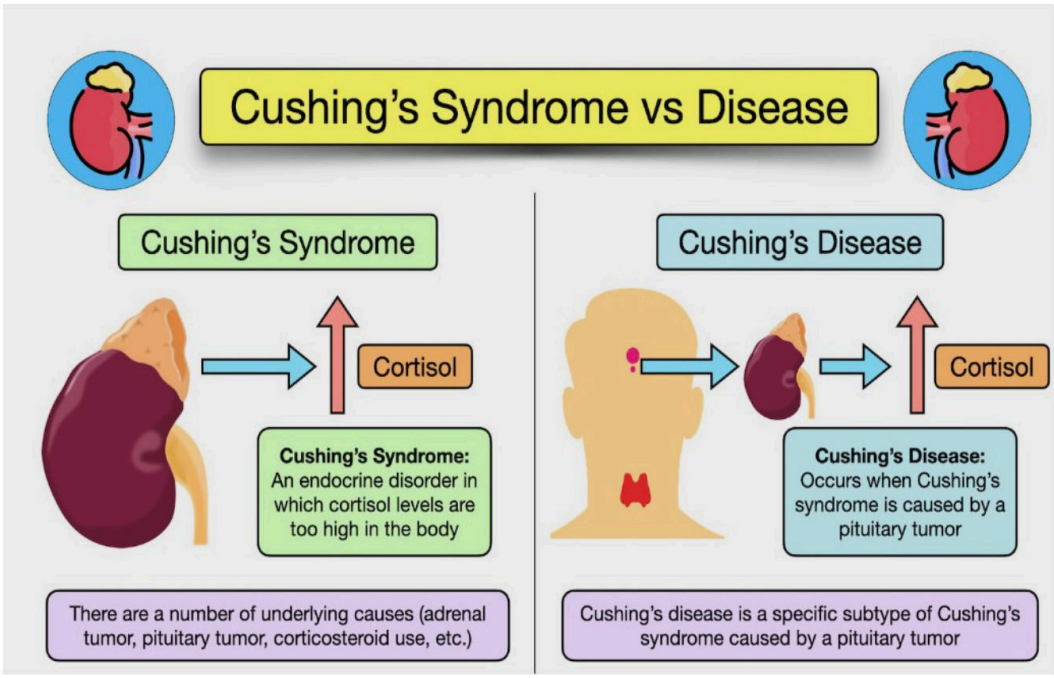
▶️ **Up next: Cushing's syndrome.**

2.9 Cushing's Syndrome: Causes (Slides 22–24) ⭐

Type	Share	Detail
A. Endogenous	Rare (**1/million/year**)	
1. ⭐ **Pituitary = Cushing DISEASE**	**75%** of endogenous	**80% from a basophil microadenoma** secreting ACTH; the rest may be hypothalamic
2. **Adrenal Cushing**	**20%**	**Autonomous cortisol** from an adenoma, carcinoma or hyperplasia
3. **Ectopic ACTH**	(~5%)	**Bronchogenic carcinoma, pulmonary carcinoid**; thymoma; ovarian or pancreatic carcinoma
B. Exogenous ("Cushingoid")	⭐ **Commonest overall** *[added]*	Prolonged **corticosteroid** therapy

Cushing's syndrome vs disease (slide 24):

- **Syndrome** = hypercortisolism from **any** cause.
- **Disease** = specifically a **pituitary ACTH-secreting tumour**.



Slide 24: Cushing's syndrome vs Cushing's disease

2.10 Clinical Picture (Slides 25–29) ⭐

Group	Features
A. Fat redistribution	⭐ **Moon face** (bloated cheeks) · ⭐ **buffalo hump** (interscapular) · **truncal obesity** (breasts, abdomen) with **hollow buttocks and thin limbs** ("**lemon on matchsticks**")
B. Protein catabolism	**Muscle wasting → proximal myopathy** · ⭐ **osteoporosis** (kyphosis, height loss, **pathological fractures**) · **bruises and purpura** · **delayed wound healing** · ⭐ **striae rubrae** (purple, wide, over the breasts, abdomen and thighs)
C. Carbohydrate	**Hyperglycaemia**; **DM in 15%** (insulin-resistant "**steroid diabetes**")
D. Fluid and electrolytes	**Na retention → HTN** · **hypokalaemia → polyuria, alkalosis**

Group	Features
E. General	**Plethoric face** (polycythaemia) · ⭐ **psychiatric**: **depression** commonest · in children: **obesity + stunted growth**
F. Sexual	**Women** (adrenal androgen excess): **amenorrhoea, acne, hirsutism** · **Men**: ↓ libido, impotence
G. Clues to the cause	⭐ **Virilisation → adrenal carcinoma** · ⭐ **hypokalaemic alkalosis, myopathy and hyperpigmentation → ECTOPIC** (classic cushingoid features often absent) · **children → adrenal carcinoma** · **adult men, especially smokers → ectopic** · **women of childbearing age → pituitary**

A. Abnormal deposition of fat in certain sites :

- In the face→ becomes rounded with bloated cheeks (Moon face).
- In the interscapular region → (Buffalo hump).
- In the breasts & abdomen (Trunkal obesity), but the buttocks are hollow with thin limbs (lemon on match sticks).



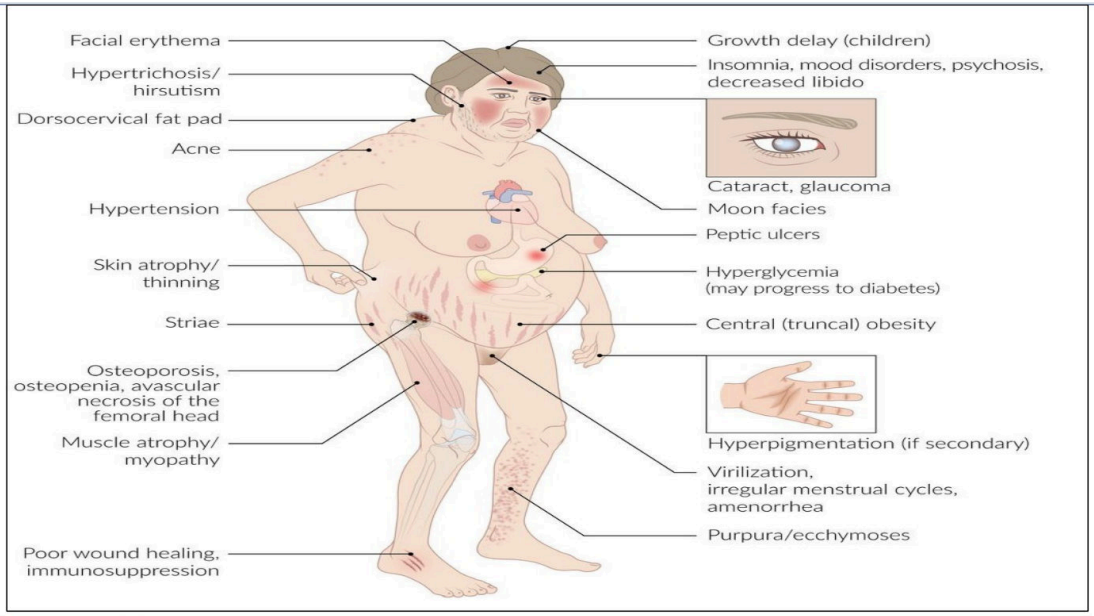
Slide 25: moon face, buffalo hump, truncal obesity

B. Disturbances in protein metabolism:

- ↑ Protein catabolism →muscle wasting & weakness even proximal myopatny).
- Osteoporosis :
  - o Kyphosis, shortening
  - o Pathological fractures.
- Bruises & purpura (due to ↓ support of blood vessels).
- Delayed healing of wounds.
- Stria rubra : Due to rupture of weakened s c. collagen fibers (together with fat mobilization) →purplish lines in the skin around the breasts, abdomen & thighs.



Slide 26: purple abdominal striae



Student pearl

"CUSHINGOID" is the acronym for side effects of corticosteroids: Cataracts, Ulcers, Striae/Skin thinning, Hypertension/Hirsutism/Hyperglycemia, Infections, Necrosis (of the femoral head), Glucose elevation, Osteoporosis/Obesity, Immunosuppression, Depression/Diabetes.

Slide 29: clinical features of Cushing's syndrome

⭐ **Mnemonic (slide 29): CUSHINGOID** = **C**ataracts · **U**lcers · **S**triae/**S**kin thinning · **H**ypertension/**H**irsutism/**H**yperglycaemia · **I**nfections · **N**ecrosis (femoral head) · **G**lucose elevation · **O**steoporosis/**O**besity · **I**mmunosuppression · **D**epression/**D**iabetes.

[added] Most discriminating signs vs simple obesity: proximal myopathy, easy bruising, wide purple striae (> 1 cm), thin skin.

2.11 Investigations (Slides 30–33) ⭐

1. Suggestive:

- **hypokalaemic alkalosis**, hypernatraemia
- **↑ RBC, ↓ lymphocytes and eosinophils**
- **impaired glucose tolerance**
- X-ray: **osteoporosis**, unsuspected fractures

2. Confirmatory (hypercortisolism):

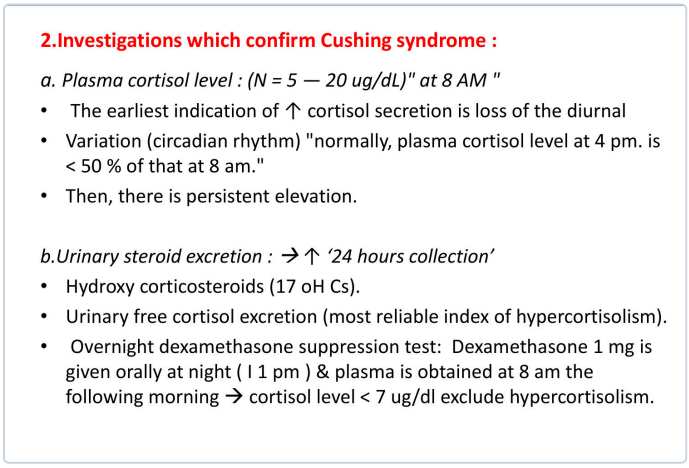
Test	Detail
Plasma cortisol	N **5–20 µg/dL at 8 am**. ⭐ **The earliest sign is loss of diurnal variation** (normally 4 pm cortisol < 50% of 8 am), then persistent elevation
⭐ **24-h urinary free cortisol**	**Most reliable index of hypercortisolism** (± 17-OH-corticosteroids)
⭐ **Overnight dexamethasone suppression**	**1 mg at 11 pm** → 8 am cortisol **< 7 µg/dL excludes** hypercortisolism (*[added]* modern cut-off < 1.8 µg/dL)

3. Differentiate the three forms:

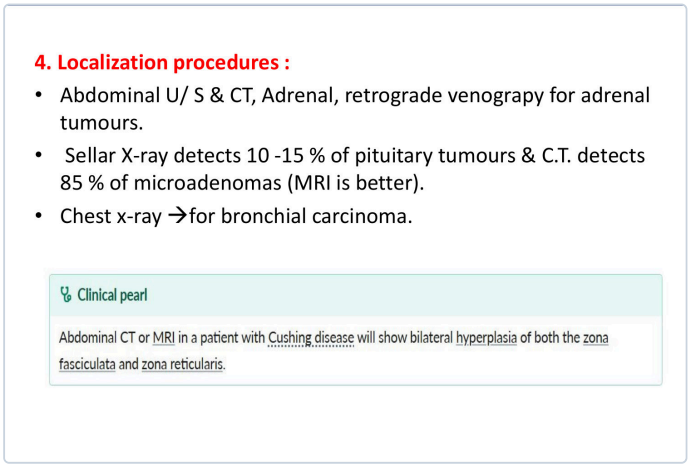
Test	Pituitary	Adrenal	Ectopic
⭐ **Plasma ACTH**	**High**	**Low or absent**	**Very high**
⭐ **High-dose dexamethasone** (2 mg 6-hourly × 2 days)	**≥ 50% suppression** of cortisol / 17-OHCS	**No effect**	**No effect**

4. Localisation:

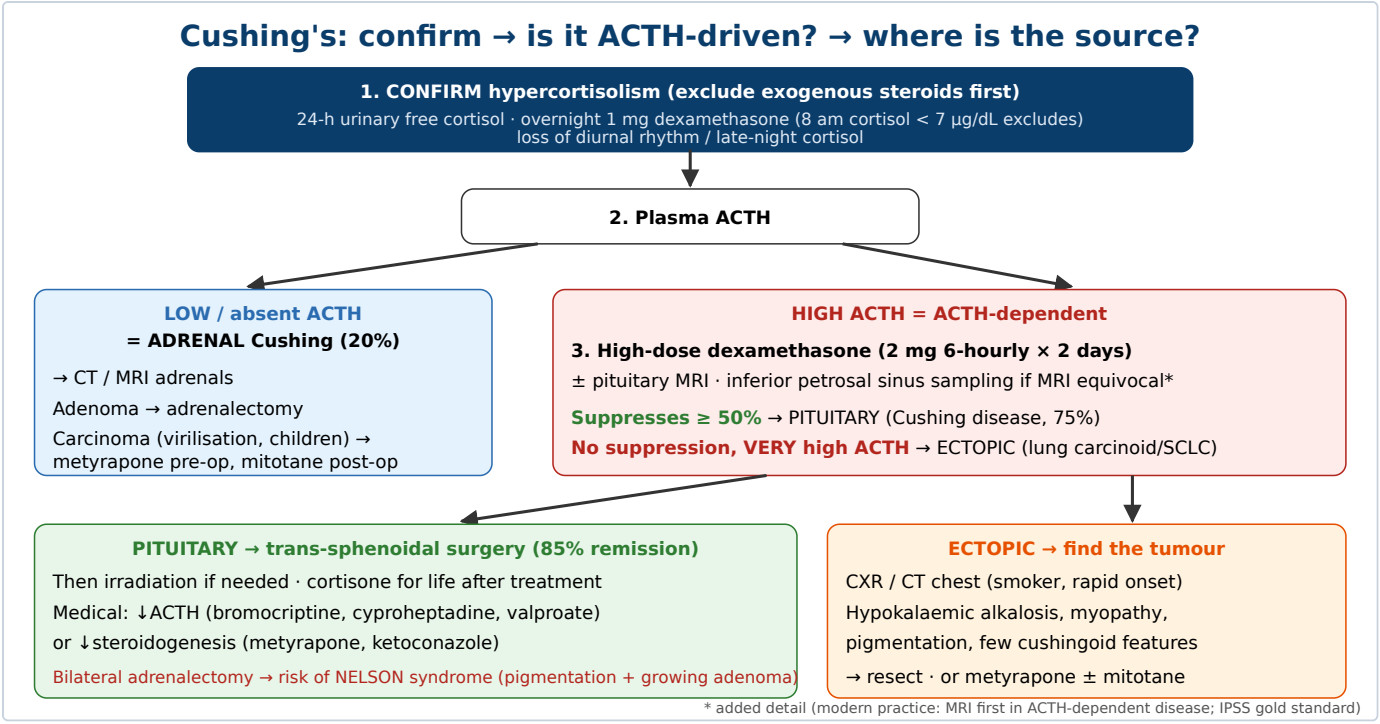
- **Adrenal:** abdominal **US and CT**, adrenal venography.
- **Pituitary:** sellar X-ray detects only **10–15%**; **CT detects 85% of microadenomas**; **MRI is better**.
- **Ectopic: CXR** for bronchial carcinoma.
- **Pearl (slide 33):** in **Cushing disease**, CT/MRI shows **bilateral hyperplasia** of the zona fasciculata and reticularis (ACTH drive).



Slide 31: confirmatory tests



Slide 33: localisation procedures



Added diagram: Cushing's work-up (confirm → ACTH → source → treatment)

2.12 Differential Diagnosis (Slide 34)

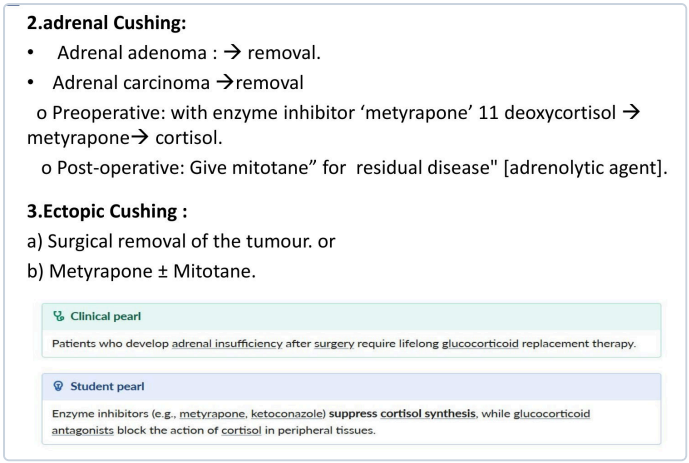
- **Simple obesity, essential HTN and DM** (alone or combined): obesity is **generalised**; cortisol and urinary end products are **normal**.
- **Other causes of obesity:** myxoedema, Fröhlich's syndrome. Other causes of hirsutism and osteoporosis.
- ⭐ **Oral oestrogen-containing contraceptives:** ↑ **cortisol-binding globulin** → ↑ total plasma cortisol (not a true excess). *[added] Use urinary free cortisol.*
- **Liver cell failure:** impaired cortisol metabolism → Cushing-like features with raised cortisol, but liver disease dominates.
- ⭐ **Pseudo-Cushing's: chronic alcoholism, severe illness, depression, obesity** → mild ↑ cortisol and **absent diurnal variation**.

2.13 Treatment (Slides 35–37) ⭐

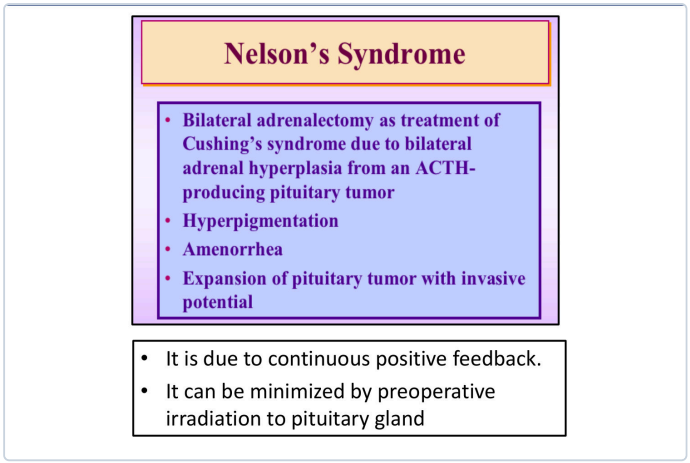
Cause	Treatment
⭐ **Pituitary Cushing**	**(a) Trans-sphenoidal hypophysectomy**: **85% remission**. **(b) Irradiation** (proton beams, alpha particles). After treatment: **cortisone for life** (if insufficient). **(c) Medical**: ↓ ACTH release (**bromocriptine, cyproheptadine** (hypothalamic serotonin antagonist), **sodium valproate** (GABA-transaminase inhibitor)); ↓ steroidogenesis
Adrenal adenoma	**Removal**
Adrenal carcinoma	**Removal**. **Pre-op: metyrapone** (blocks 11-deoxycortisol → cortisol). **Post-op: mitotane** (adrenolytic) for residual disease
Ectopic	**Surgical removal of the tumour**, or **metyrapone ± mitotane**

Pearls:

- Patients who develop **adrenal insufficiency after surgery** need **lifelong glucocorticoid replacement**.
- **Enzyme inhibitors** (metyrapone, **ketoconazole**) **suppress cortisol synthesis**; **glucocorticoid antagonists** (*[added]* mifepristone) block cortisol action in the tissues.



Slide 36: adrenal and ectopic Cushing treatment



Slide 37: Nelson's syndrome

Nelson's syndrome (slide 37) ⭐

- **Cause: bilateral adrenalectomy** as treatment for Cushing's from bilateral adrenal hyperplasia due to an ACTH-producing pituitary tumour.
- **Features:** - **hyperpigmentation** - **amenorrhoea** - **expansion of the pituitary tumour**, with **invasive potential**
- **Mechanism: loss of cortisol negative feedback** (the slide says "continuous positive feedback").
- **Prevention: pre-operative pituitary irradiation**.

> 🔁 **Recap: Cushing's.**
>
> - **Causes:** exogenous steroids (commonest overall); endogenous: pituitary 75% (Cushing disease), adrenal 20%, ectopic (SCLC, carcinoid).
> - **Features:** moon face, buffalo hump, truncal obesity, thin limbs, proximal myopathy, osteoporosis, bruising, purple striae, HTN, DM 15%, depression, hirsutism.
> - **Clues:** virilisation → carcinoma; hypoK + pigmentation → ectopic.
> - **Confirm:** urinary free cortisol (most reliable), overnight 1 mg dex (< 7 excludes), loss of diurnal rhythm.
> - **Find the source:** ACTH (low = adrenal; high = pituitary; very high = ectopic); high-dose dex suppresses only pituitary. MRI pituitary, CT adrenal, CXR.
> - **Treatment:** trans-sphenoidal (85%), adrenalectomy, metyrapone/mitotane. Bilateral adrenalectomy → Nelson's.

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3. High-Yield Comparison Tables

Table A: Adrenal Zones

Zone	Hormone	Regulator	Excess	Deficiency
Glomerulosa	Aldosterone	Angiotensin II, K^+	**Conn's**	Addison's (1ry only)
Fasciculata	Cortisol	ACTH	**Cushing's**	Addison's / 2ry AI
Reticularis	Androgens	ACTH	Virilisation (carcinoma, CAH)	↓ axillary and pubic hair (♀)
Medulla	Catecholamines	Sympathetic	**Pheochromocytoma**	—

Table B: Hypertension + Hypokalaemia by Renin and Aldosterone

Renin	Aldosterone	Diagnosis
↓	↑	**Primary hyperaldosteronism** (Conn's adenoma, bilateral hyperplasia)
↑	↑	**Secondary**: renal artery stenosis, renin-secreting tumour, diuretics, malignant HTN, CHF, cirrhosis
↓	↓	**Pseudo**: Cushing's, CAH (11β/17α), exogenous mineralocorticoid, **liquorice**, **Liddle's** *[added]*

Table C: Cushing's Subtypes

	Pituitary	Adrenal	Ectopic
Share	75%	20%	~5%
Typical patient	**Woman of childbearing age**	**Child** (carcinoma); adult (adenoma)	**Older male smoker**
ACTH	↑	↓	↑↑
High-dose dex	**Suppresses**	No	No
Pigmentation	±	**No**	**Yes**
HypoK alkalosis	Mild	Mild	**Marked**
Clue	Gradual, classic features	**Virilisation (carcinoma)**	Rapid, weight loss, myopathy, few classic signs
Imaging	**Pituitary MRI**	**Adrenal CT**	**CXR / CT chest**
Rx	**Trans-sphenoidal**	**Adrenalectomy** (± metyrapone/mitotane)	**Resect tumour** / metyrapone ± mitotane

Table D: Dexamethasone Tests

Test	Dose	Purpose	Interpretation
Overnight (low-dose)	**1 mg at 11 pm**	**Screen / confirm**	8 am cortisol **< 7 µg/dL → excludes**
High-dose	**2 mg 6-hourly × 2 days**	**Pituitary vs others**	**≥ 50% suppression → pituitary**; no effect → adrenal/ectopic

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Aldosterone excess → oedema?	**No oedema** in Conn's (aldosterone escape); oedema appears in **2ry** states
Measure renin as is	**Stop antihypertensives first** (they alter renin)
Spironolactone for Conn's	Lecture prefers **amiloride** (no gynaecomastia/impotence); *[added]* eplerenone
Adenoma on CT = cause	Confirm lateralisation with **adrenal venous sampling** before surgery
Cushing's = pituitary	Pituitary is commonest **endogenous** cause; **exogenous steroids** are commonest overall *[added]*

Rule	Exception / nuance
Cushing's = obese, round face	**Ectopic** may lack classic features (rapid course): **hypoK + pigmentation + myopathy**
High plasma cortisol = Cushing's	**OCP** (↑ CBG), **liver failure**, **pseudo-Cushing's** (alcohol, depression, obesity)
High-dose dex suppresses cortisol	**Only pituitary** (ACTH partly feedback-sensitive)
Bilateral adrenalectomy cures Cushing disease	**Nelson's syndrome**: pigmentation + enlarging pituitary tumour
Hypokalaemia → weakness	Can cause **episodic paralysis, ileus and IGT**

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5. Clinical Correlations

- **Young hypertensive on 3 drugs with K^+ 2.8** → screen for **Conn's** (aldosterone:renin ratio).
- **Hypertensive on a thiazide with low K^+** → could be the drug, but persistent hypokalaemia off diuretics → Conn's.
- **Liquorice eater with HTN, low K^+, low renin and low aldosterone** → pseudo-hyperaldosteronism *[added]*.
- **Woman of 30 with weight gain, depression, irregular menses, purple striae, proximal weakness** → **Cushing disease** (lecture case).
- **Older male smoker, rapid weakness, weight loss, pigmentation, K^+ 2.5, ACTH very high** → **ectopic ACTH (SCLC)**.
- **Child with Cushing's features + virilisation** → **adrenal carcinoma**.
- **Asthma patient on long-term oral prednisolone with moon face** → **exogenous (iatrogenic) Cushing's**. *[added]* Don't stop abruptly (→ adrenal crisis, lecture 13).
- **Alcoholic with cushingoid look and mildly raised cortisol** → **pseudo-Cushing's**: features resolve with abstinence.

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6. Rapid Revision (Last-Day)

1. **GFR = salt, sugar, sex**: glomerulosa (aldosterone, RAAS), fasciculata (cortisol, ACTH), reticularis (androgens, ACTH); medulla = catecholamines.
2. Cortisol: gluconeogenesis, protein catabolism, fat redistribution, Na retention/K loss, ↑RBC, ↓eosinophils and lymphocytes, anti-inflammatory.
3. **Conn's:** adenoma **60%**, bilateral hyperplasia **40%**, carcinoma rare.
4. **HTN + hypokalaemia** (weakness, paralysis, ileus, arrhythmia, IGT, polyuria) + **metabolic alkalosis** (tetany) + **NO oedema** (aldosterone escape).
5. **↑ aldosterone, ↓ renin**, **ARR > 20**; **stop antihypertensives** before renin.
6. Confirm: **oral Na load** (urine aldosterone > 12 µg/day) or **saline infusion** (aldosterone > 10 ng/dL); fludrocortisone and captopril tests.
7. Then **adrenal CT → adrenal venous sampling** → unilateral: **adrenalectomy**; bilateral: **MRA / amiloride** (amiloride 40 mg avoids gynaecomastia).
8. **2ry hyperaldosteronism:** ↑ renin (RAS, HF, nephrotic, cirrhosis). **Pseudo:** ↓ renin, ↓ aldosterone (Cushing's, CAH, exogenous MC, Liddle's).
9. **Cushing's: pituitary 75%** (basophil microadenoma), **adrenal 20%**, **ectopic** (bronchogenic carcinoma, carcinoid, thymoma); **exogenous** = commonest overall.
10. **Moon face, buffalo hump, truncal obesity, thin limbs, proximal myopathy, osteoporosis, bruises, purple striae, HTN, DM 15%, plethora, depression, hirsutism/amenorrhoea.**

11. **Virilisation → carcinoma; hypoK + pigmentation + myopathy → ectopic; children → carcinoma; smokers → ectopic; women of childbearing age → pituitary.**
12. Confirm: **loss of diurnal rhythm** (earliest), **24-h urinary free cortisol** (most reliable), **overnight 1 mg dex** (< 7 excludes).
13. Source: **ACTH** low (adrenal) / high (pituitary) / very high (ectopic); **high-dose dex** (2 mg 6-hourly × 2 days) suppresses **pituitary only**.
14. Localise: adrenal CT, pituitary **MRI** (CT 85%; X-ray 10–15%), CXR.
15. DDx: simple obesity, **OCP (↑ CBG)**, liver failure, **pseudo-Cushing's** (alcohol, depression).
16. Treatment: **trans-sphenoidal (85%)**, irradiation, bromocriptine/cyproheptadine/valproate; adrenal → removal (**metyrapone pre-op, mitotane post-op**); ectopic → resect or metyrapone ± mitotane. Lifelong steroids if insufficiency develops.
17. **Nelson's:** bilateral adrenalectomy → pigmentation + amenorrhoea + expanding invasive pituitary tumour; prevent with pre-op pituitary irradiation.

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7. Exam-Style Questions

Q1 (lecture case 1, slides 38–39). A 32-year-old woman has weight gain, weakness, irregular menses, depression, difficulty rising from a chair, HTN, plethoric face, central obesity and violaceous striae. Midnight cortisol is high twice, 24-h urinary cortisol is high, and **ACTH is elevated (136 pg/mL)**. Next step in evaluation? A. MRI head with contrast · B. Inferior petrosal sinus sampling · C. High-dose dexamethasone suppression test · D. Low-dose dexamethasone suppression test · E. CT abdomen · F. Ketoconazole · G. Bilateral adrenalectomy

> **Answer: A. MRI of the head with contrast.** *[reasoning added; key not on slides]*
>
> - Hypercortisolism is already **confirmed** (two late-night cortisols + urinary free cortisol), so **D is redundant**.
> - **ACTH is high = ACTH-dependent**. A woman of childbearing age with gradual classic features most likely has **Cushing disease**. Current guidance images the **pituitary first** (MRI). If MRI is negative or equivocal → **IPSS (B)**.
> - ⚠️ **Lecture caveat:** this lecture's algorithm (slide 32) uses the **high-dose dexamethasone test (C)** to separate pituitary from ectopic. If the examiner follows the slides, **C** is defensible. Know both.
> - **CT abdomen (E)** is for **ACTH-independent** (low ACTH) disease.
> - F and G are treatment, not evaluation.

Q2 (lecture case 2, slides 40–41). A 42-year-old hypertensive man (BP 150/102) has **K^+ 2.8**. Off enalapril: **aldosterone 35 ng/dL, renin 0.4**, and oral salt loading **fails to suppress** aldosterone. Next step? a) Adrenal venous sampling · b) Spironolactone · c) Fludrocortisone suppression test · **d) Adrenal CT scan** · e) Adrenalectomy · f) MRI abdomen

> **Answer: d. Adrenal CT.** *[reasoning added]*
>
> - **Primary hyperaldosteronism is confirmed** (high ARR + non-suppression).
> - Next is **subtype identification with adrenal CT** (slide 21), then **adrenal venous sampling (a)** to lateralise before surgery.
> - **c** is another confirmatory test, which is no longer needed.
> - **e** comes before lateralisation, which is wrong.
> - **b** is treatment for bilateral disease or poor surgical candidates.

Q3. Which finding best suggests **ectopic ACTH** rather than pituitary Cushing's? A. Purple striae · B. Buffalo hump · **C. Hypokalaemic alkalosis with hyperpigmentation and rapid myopathy** · D. Suppression on high-dose dexamethasone · E. Woman aged 30

> **Answer: C** (slide 28).
> - D and E favour **pituitary**.

Q4. A hypertensive man has K^+ 2.9, ↑ aldosterone and ↑ renin. Most likely? A. Conn's adenoma · **B. Renal artery stenosis (secondary hyperaldosteronism)** · C. Liddle's syndrome · D. Exogenous mineralocorticoid · E. 11β-hydroxylase deficiency

> **Answer: B.**
> - ↑ renin + ↑ aldosterone = **secondary** (slides 20–21).
> - **A** has ↓ renin.
> - **C, D and E** have ↓ renin and ↓ aldosterone.

Q5. Years after bilateral adrenalectomy for Cushing disease, a patient develops deep skin pigmentation, headaches and a bitemporal field defect. Diagnosis? A. Addison's disease · **B. Nelson's syndrome** · C. Ectopic ACTH · D. Haemochromatosis · E. Adrenal crisis

> **Answer: B** (slide 37).
> - Loss of cortisol feedback → **ACTH-secreting pituitary tumour enlarges** (field defect) with **hyperpigmentation**.

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8. Exam Pitfalls & Traps

- **Conn's has NO oedema.** Aldosterone escape is the reason, and it is an MCQ favourite.
- **Stop antihypertensives before measuring renin.** ACEI/ARBs and diuretics ↑ renin; β-blockers ↓ renin *[added]*.
- **Slide 15 lists "hypernatraemia"** in Conn's. Na^+ is usually **high-normal** because of aldosterone escape *[added]*.
- **Slide 18 prefers amiloride** over spironolactone (anti-androgen effects). Standard guidelines use **spironolactone/eplerenone** for bilateral hyperplasia *[added]*.
- **Slide 19 lists Bartter's** among hypertensive K-losing states. Bartter's is **normotensive** (unlike Liddle's) *[added]*.
- **Overnight dex cut-off:** the slide says **< 7 µg/dL** excludes. Modern guidelines use **< 1.8 µg/dL** (more sensitive) *[added]*. Use the lecture value in the lecture exam.
- **Next step after confirmed ACTH-dependent Cushing's:** the lecture uses **high-dose dex**; current practice is **pituitary MRI**, then IPSS. Read the stem (slides vs AMBOSS-style).
- **"Cushing's disease" ≠ "Cushing's syndrome".** Disease = pituitary only.
- **OCP users** have ↑ total cortisol from ↑ CBG. Use **urinary free cortisol**.
- **Nelson's mechanism:** the slide says "continuous positive feedback"; strictly it is **loss of negative feedback** from cortisol.
- **Metyrapone** blocks **11β-hydroxylase** (11-deoxycortisol → cortisol). **Mitotane** is **adrenolytic**.